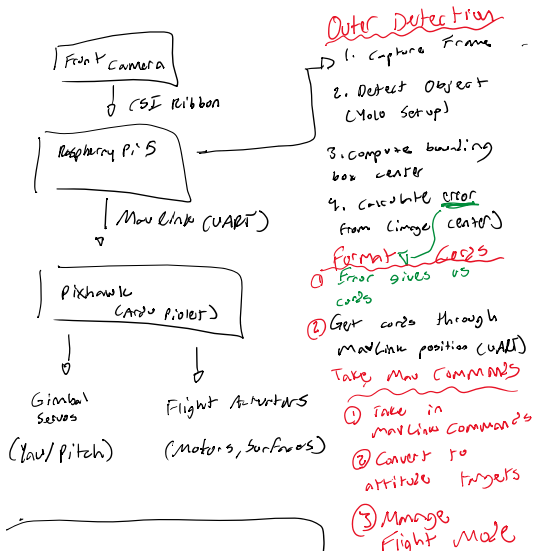




Test Ardu w/o VTOL
Ardu Pilot + SITL

Learn

- sim-vehicle.py
- mission planner
- py mavlink to TCP port
- aircraft your response

3 Main Components

① Vision / AI subsystem (R PI)

↳ Run Yolo

↳ Detect stationary targets

↳ Track Moving Targets

↳ Compute Error Values

① open CV

Image cap, Color Spaces, Bounding Boxes, Drawing Overlays

② Yolo (pretrained)

Inference on PI, confidence thresholding, bounding box extraction, optimize frame rate, resize frames, use Yolo 'nano' model

② MAVLINK communication

↳ Connect Pi to Pixhawk

↳ Send Commands

↳ Read telemetry

↳ Switch Flight Modes

① Learn MavLink

heartbeat, system ID, component ID, message types

② py MavLink (messaging to Ardu Pilot)

connect via serial

MAP

Week 1

- Install YOLO
- Detect stationary object
- Print bounding box + error

Week 2

- Learn MAVLink
- Connect to SITL
- Send mode changes
- Send yaw commands

Week 3

- Integrate detection → yaw
- Add thresholding
- Add failsafe logic

Week 4

- Add gimbal control logic
- Implement 2-layer control (gimbal first)

Week 5

- Bench integration with Pixhawk
- Full dry run
- Prepare for airframe integration

Parts List

Category	Approx Cost
Raspberry Pi 5 Essentials Kit	~\$189
Arducam Camera	~\$57
5V 5A UBEC	~\$7
JST Serial Cable	~\$8-12
2-Axis Gimbal	~\$12
SG90 Servo Pair	~\$10-12
Cables/Hardware	~\$20-30
XT60 Splitter	~\$8

1) Companion Computer (AI + Vision Processing)

Raspberry Pi 5 - 8GB Starter Kit

Raspberry Pi 5 Essentials Kit - 8GB

Includes:

- Raspberry Pi 5 8GB
- Official USB-C power supply
- 64 GB microSD card
- Case & cables (usually)

Purpose:

Runs OpenCV + YOLO + MAVLink interface code — the "brain" that detects targets and sends commands.

Why this specific part:

- More memory for AI models
- Better performance than Pi 4
- Supports CSI camera

2) Vision System (Camera)

Arducam Wide Camera Module 3 for Raspberry Pi

Arducam Wide Camera Module 3 for Raspberry Pi

Purpose:

Provides low-latency high-resolution video feed to the Pi for target detection.

Why this specific part:

- Uses CSI interface (fast)
- Compatible with Raspberry Pi 5
- Autofocus

CSI (Camera Serial Interface) is required for performance — USB webcams are too slow/laggy for real-time AI.

3) Power Regulation

5 V 5 A UBEC Voltage Regulator

5V 5A UBEC Mini Voltage Regulator

Purpose:

Provides a clean 5 V power source from your drone's main battery to power the Raspberry Pi safely.

Why this part:

- Raspberry Pi must not be powered directly from Pixhawk 5 V rail
- 5 A ensures enough headroom for camera and processing load

Handwritten scribble on the left margin.

Handwritten scribble on the left margin.

Handwritten scribble on the left margin.

② py Mav Link (Messaging to Ardu Pilot)

connect via serial

send

↳ guided mode

↳ velocity commands

↳ mount control for (gimbal)

③ Ardu Modes

Guided, Loiter, Auto, VTOL modes

How we send?

velocity VED commands

Yaw rate commands

Position targets

Parameter Setup

↳ Enable companion computer

SET SERIALx — Protocol to MAVLink

SET Baud rate

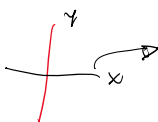
Gimbal mount config

MNT_TYPE

MNT_RC_IN_PAN

MNT_RC_IN_TILT

} MAV Link
control
messages



move left right

- Raspberry Pi must not be powered directly from Pixhawk 5 V rail
- 5 A ensures enough headroom for camera and processing load

4) Pixhawk → Companion Computer Interface Cable

JST-GH 6-Pin to UART Cable

JST-GH 6-Pin to 4-Pin UART Cable

Purpose:

Connects Pixhawk's TELEM2 port to the Raspberry Pi's UART port for MAVLink communication.

5) Camera Pan/Tilt Mount

Dual Axle 2-Axis Servo Gimbal Pan/Tilt Mount

2-Axis Servo Pan/Tilt Camera Mount

Purpose:

Allows the camera to rotate up/down + left/right (approx 180°) for ground scanning.

Why essential:

- Enables the camera to follow targets without the aircraft needing to yaw
- Important for smoother tracking

6) Servos for Gimbal

TowerPro SG90 Micro Servos (x2)

TowerPro SG90 Micro Servo (2 Pack)

Purpose:

Provides motion for the 2-axis pan/tilt mount.

Why this part:

- Small, lightweight, inexpensive
- Works well with Pixhawk AUX PWM outputs

7) Accessory Cables & Connectors

These items ensure you can *actually* wire and mount everything:

JST-GH Cable Kit

– Necessary for wiring from Pixhawk to Pi and from servos to Pixhawk AUX.

Servo Extension Cables

– To route servos to Pixhawk AUX outputs.

Heat Shrink Tubing / Zip Ties / Fasteners

– To secure wiring and protect connections.
(Kinds can be generic drone wiring accessory packs — not a single specific SKU, but essential.)

8) Power Distribution & Battery Interface

► XT60 Splitter Cable

– Allows safe power feed from the drone battery to UBEC + other electronics.

SOFTWARE & CABLING (Necessary for Integration)

MAVLink Libraries (DroneKit.py /

pymavlink)

ArduPilot set up on Pixhawk (ArduPlane / VTOL firmware)

Python 3 + OpenCV + YOLO

Serial/UART header enabled on Pi

(Software items are necessary but do not count toward parts cost.)

X

